

Metropolitan University

Department of Computer Science & Engineering

Class Test-01; Batch: 62(B)

Course: MAT 135: Matrices, Complex Variable & Fourier Analysis

Time: 1 hour

Marks:20

(Attempt all **four** questions)

Q.1. Solve the following system of linear equation with the help of Ad-joint method- (5)

$$x + y + z = 6$$

$$5x - y + 2z = 9$$

$$3x + 6y - 5z = 0$$

Q.2. Find the inverse of $A = \begin{pmatrix} 1 & 3 & -2 \\ -3 & 0 & -5 \\ 2 & 5 & 0 \end{pmatrix}$ by row canonical method. (5)

Q.3. Solve the following system by using Gaussian Elimination method- (5)

$$x + 4y + 9z = 16$$

$$2x + y + z = 10$$

$$3x + 2y + 3z = 18 \quad (\text{Determine the nature of the solution})$$

Q.4. Solve the following system by using Gauss Jordan method- (5)

$$x + 2y + 3z = 14$$

$$2x + 3y + 4z = 20$$

$$3x + 4y + 6z = 33 \quad (\text{Classify the solution type})$$

Metropolitan University

Department of CSE

CT-02; Batch- 62(B)

Course: MC&FA

Time: 1 hour

Marks: 20

(Attempt All Questions)

Q.1. Find the eigenvalues and eigenvectors of $A = \begin{pmatrix} 8 & -8 & -2 \\ 4 & -3 & -2 \\ 3 & -4 & 1 \end{pmatrix}$ (8)

Q.2. Solve the following system by using LU Factorization method- (8)

$$x + 2y + 3z = 9$$

$$4x + 5y + 6z = 24$$

$$3x + y - 2z = 4$$

Q.3. Show that $\left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|}$ (4)